

WordPress Website Hosting on RHEL EC2

Description

Instructions:

1. Install the necessary packages, which include MySQL 10.6, PHP version 8 or higher, and Nginx on your Linux server.
2. Create a user's home directory under the '/home' directory. You should set the root directory for the website to be '/home/username/website/public.'
3. Configure SFTP (Secure File Transfer Protocol) access for the user to ensure secure file management.
4. Configure phpmyadmin access for the user to ensure secure database management.
5. Install a free SSL certificate to enable secure HTTPS access to your website.
6. Once the website is hosted, proceed to the WordPress dashboard to make necessary configurations and customizations.
7. Write a personal blog post about yourself using the WordPress content management system.

Please share the following credentials after you finish the test.

Please ensure that you document each step of the process and any configurations made. You may be asked to provide a brief report detailing the steps you took to complete this task.

1. Installation of Required Packages

An AWS EC2 instance running **RHEL 10.2** was used as the Linux server.

The required software was installed and configured:

- Nginx – Web server
- PHP 8.3 – PHP runtime
- PHP-FPM – PHP processing service
- MariaDB 11.8.8 – Database server

The assignment mentioned MySQL 10.6. Since MySQL does not have a 10.6 release, MariaDB was used as the compatible database server. The latest available MariaDB version from the RHEL repository was **11.8.8**.

The installed versions were verified using:

```
nginx -v
php -v
mariadb --version
```

```
[ec2-user@ip-172-31-7-25 ~]$ nginx -v
nginx version: nginx/1.26.3
[ec2-user@ip-172-31-7-25 ~]$ php -v
PHP 8.3.33 (cli) (built: Jul 28 2026 17:56:10) (NTS gcc x86_64)
Copyright (c) The PHP Group
Zend Engine v4.3.33, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.33, Copyright (c), by Zend Technologies
[ec2-user@ip-172-31-7-25 ~]$ mariadb --version
mariadb from 11.8.8-MariaDB, client 15.2 for Linux (x86_64) using EditLine wrapper
[ec2-user@ip-172-31-7-25 ~]$ |
```

2. Creating the Website User and Directory

A separate Linux user named **sanjid** was created for managing the website.

The website directory was created according to the required structure:

```
/home/sanjid/mywebsite/public
```

The directory structure was:

```
/home/
├── sanjid/
│   ├── mywebsite/
│   │   └── public/
```

The public directory was configured as the document root for the website.

The WordPress files were placed inside this directory.

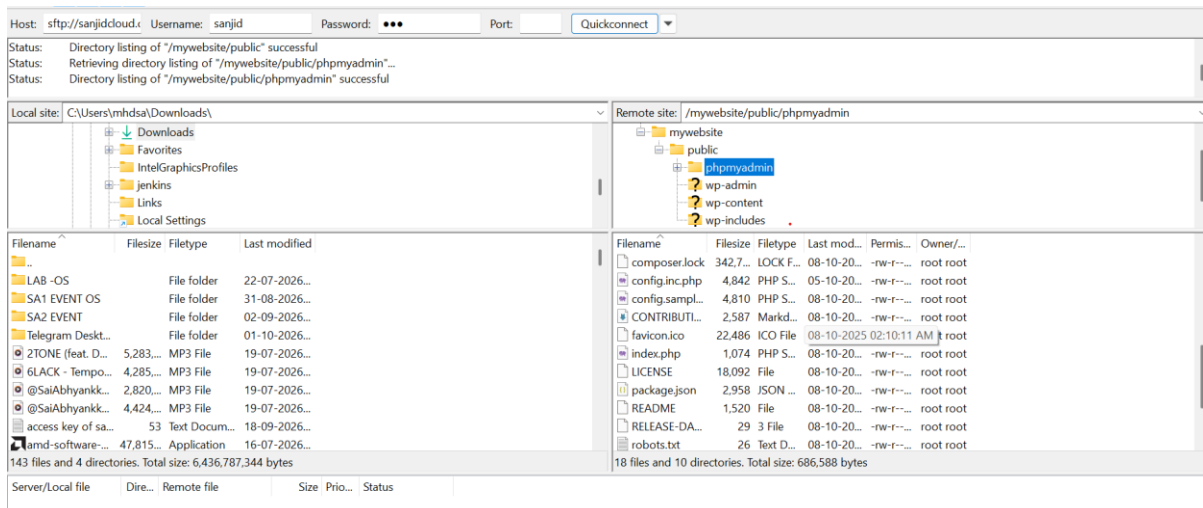
The final website directory also contained the phpMyAdmin directory:

```
public/
├── index.php
```

```

├─ wp-admin/
├─ wp-content/
├─ wp-includes/
└─ phpmyadmin/
    └─ index.php

```



3. Nginx and PHP-FPM Configuration

Nginx was configured to serve the website from:

/home/sanjid/mywebsite/public

PHP-FPM was configured as the PHP processing service.

The PHP-FPM socket used by Nginx was:

/run/php-fpm/www.sock

Nginx was configured to process PHP files through PHP-FPM.

The Nginx configuration was tested using:

```
sudo nginx -t
```

The configuration was successfully tested and Nginx was reloaded.

4. SELinux Configuration

Since RHEL uses SELinux, the correct SELinux context was assigned to the website directory.

The website files were assigned the `httpd_sys_content_t` context and the context was restored using:

```
sudo restorecon -Rv /home/sanjid/mywebsite/public
```

This allowed Nginx to access and serve the website files correctly.

5. SFTP Configuration

SFTP access was configured for the `sanjid` user.

The user was restricted to SFTP access and was chrooted to its home directory.

The SFTP configuration included:

```
Match User sanjid
    PasswordAuthentication yes
    KbdInteractiveAuthentication yes
    ChrootDirectory %h
    ForceCommand internal-sftp
    X11Forwarding no
    AllowTcpForwarding no
```

The SSH configuration was tested and the SSH service was restarted.

SFTP access was successfully configured for secure website file management.

The user's SFTP root corresponded to:

```
/mywebsite/public
```

which corresponds to:

```
/home/sanjid/mywebsite/public
```

on the server.

```

minusa@HP-DESKTOP-MINQW04 ~
$ sftp sanjid@sanjidcloud.duckdns.org
The authenticity of host 'sanjidcloud.duckdns.org (15.207.248.33)' can't be established.
ED25519 key fingerprint is: SHA256:8G1G756qrwHkHtLtvNG6yC9Q+idFCphZ+XylDpsiHM4
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:124: ec2-15-207-248-33.ap-south-1.compute.amazonaws.com
  ~/.ssh/known_hosts:127: 15.207.248.33
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'sanjidcloud.duckdns.org' (ED25519) to the list of known hosts.
(sanjid@sanjidcloud.duckdns.org) Password:
Connected to sanjidcloud.duckdns.org.
sftp> ls
mywebsite
sftp> cd mywebsite/
sftp> p
progress put      pwd
sftp> cd public/
sftp> pwd
Remote working directory: /mywebsite/public
sftp> ls
index.php          license.txt         readme.html         wp-activate.php
wp-admin           wp-blog-header.php wp-comments-post.php wp-config-sample.php
wp-config.php      wp-content         wp-cron.php        wp-includes
wp-links-opml.php  wp-load.php        wp-login.php       wp-mail.php
wp-settings.php    wp-signup.php      wp-trackback.php   xmlrpc.php

```

6. MariaDB Database Configuration

A MariaDB database was created for WordPress.

The database details were:

Database: wordpress

Database User: wpuser

Database Host: localhost

The wpuser account was granted privileges on the wordpress database.

The database was then used for the WordPress installation.



7. phpMyAdmin Installation and Configuration

phpMyAdmin was installed manually because the required package was not available in the RHEL repositories.

phpMyAdmin version **5.2.3** was installed and configured with Nginx and PHP-FPM.

The phpMyAdmin files were placed under:

`/home/sanjid/mywebsite/public/phpmyadmin`

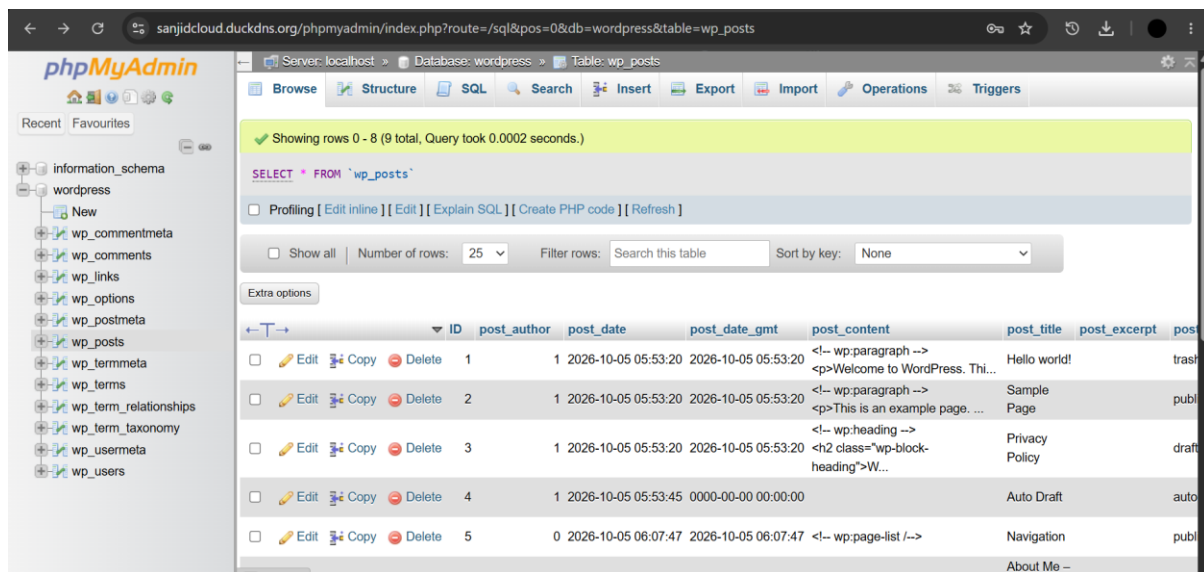
The directory contained the phpMyAdmin `index.php` file.

Nginx was configured to serve phpMyAdmin from this directory.

phpMyAdmin was successfully accessed through:

<https://sanjidcloud.duckdns.org/phpmyadmin/>

The wordpress database was visible after logging in with the database credentials.



8. WordPress Installation

WordPress was downloaded and installed inside:

`/home/sanjid/mywebsite/public`

The WordPress configuration file was created as:

`/home/sanjid/mywebsite/public/wp-config.php`

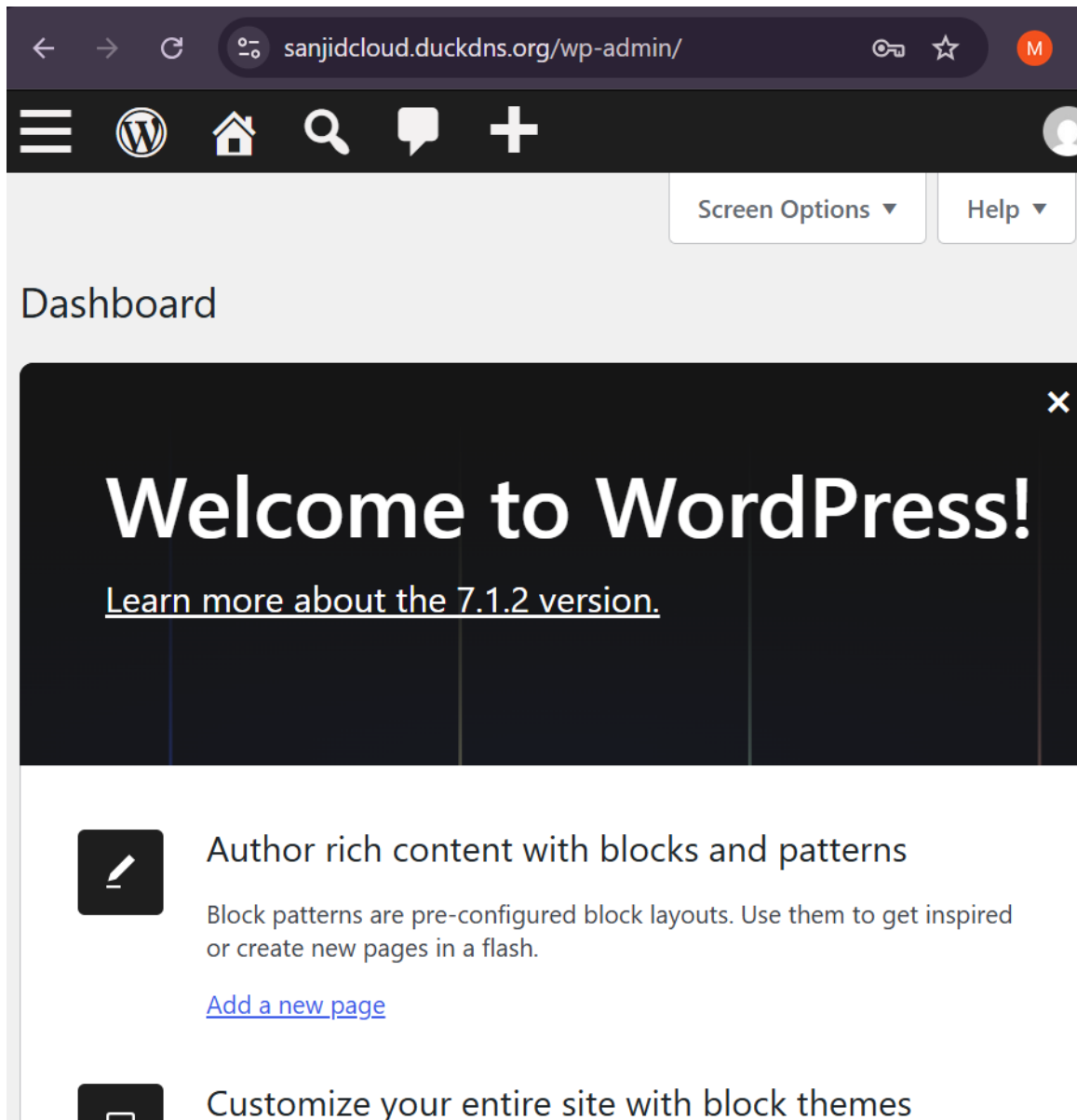
WordPress was connected to the MariaDB database using:

Database: `wordpress`

Username: `wpuser`

Host: `localhost`

The WordPress installation was completed successfully.



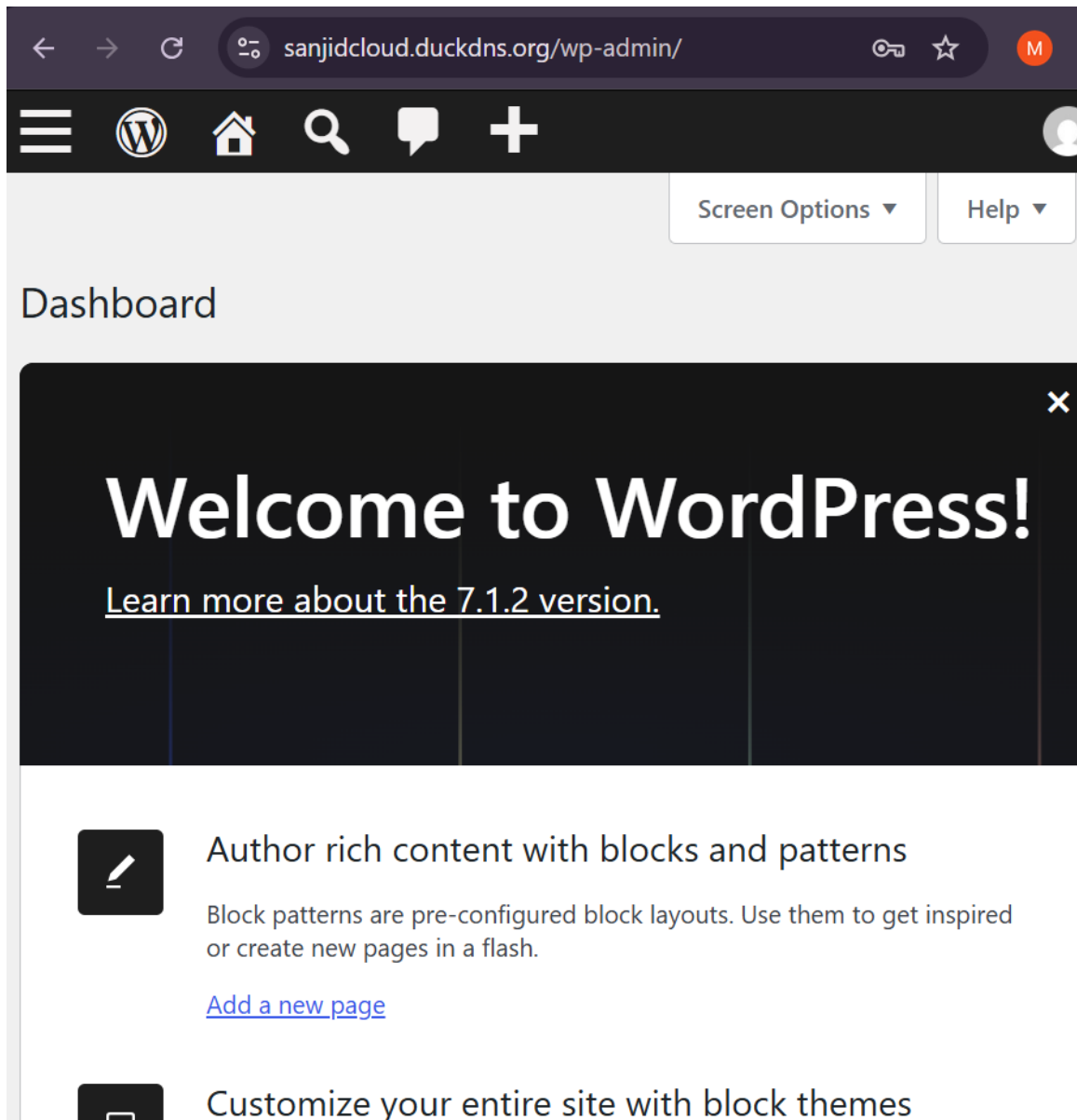
9. Domain and DNS Configuration

A free DuckDNS hostname was used for the website:

`sanjidcloud.duckdns.org`

The DuckDNS record was configured to point to the EC2 server's public IP address.

The website was then accessible through the hostname instead of directly using the EC2 IP address.



10. SSL Certificate and HTTPS Configuration

A free **Let's Encrypt SSL certificate** was installed using Certbot.

The certificate was configured for:

`sanjidcloud.duckdns.org`

Nginx was automatically configured to use the SSL certificate.

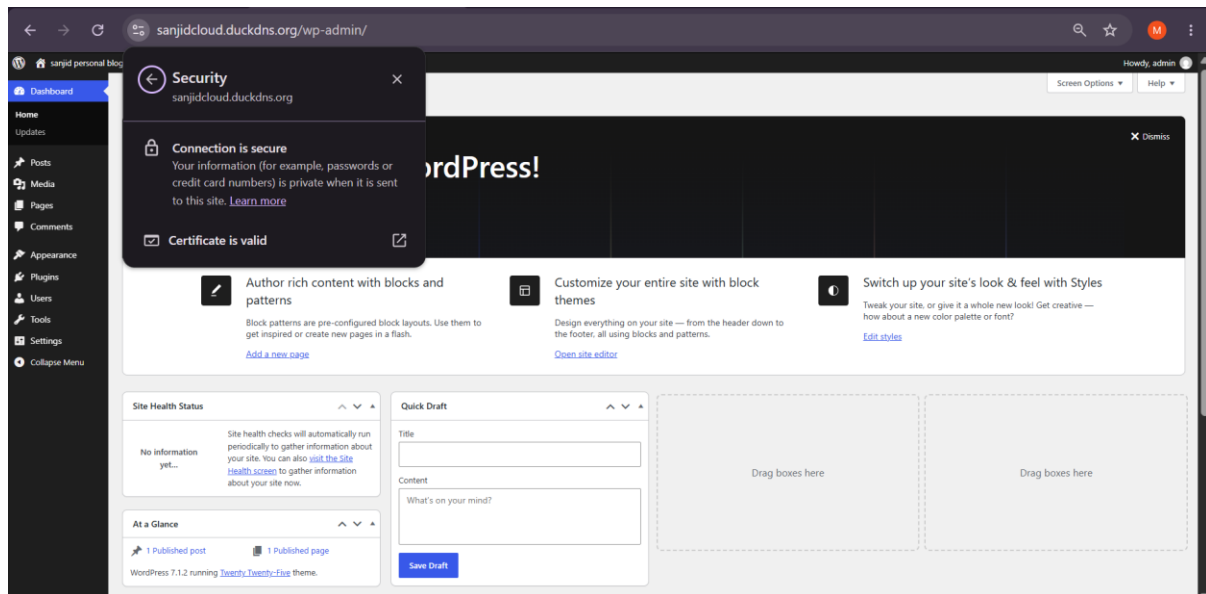
The website was successfully accessed using:

<https://sanjidcloud.duckdns.org/>

HTTPS was also used for phpMyAdmin:

<https://sanjidcloud.duckdns.org/phpmyadmin/>

The browser displayed the secure connection/lock icon, confirming HTTPS access.



11. WordPress Dashboard Configuration

After completing the WordPress installation, the WordPress dashboard was accessed through:

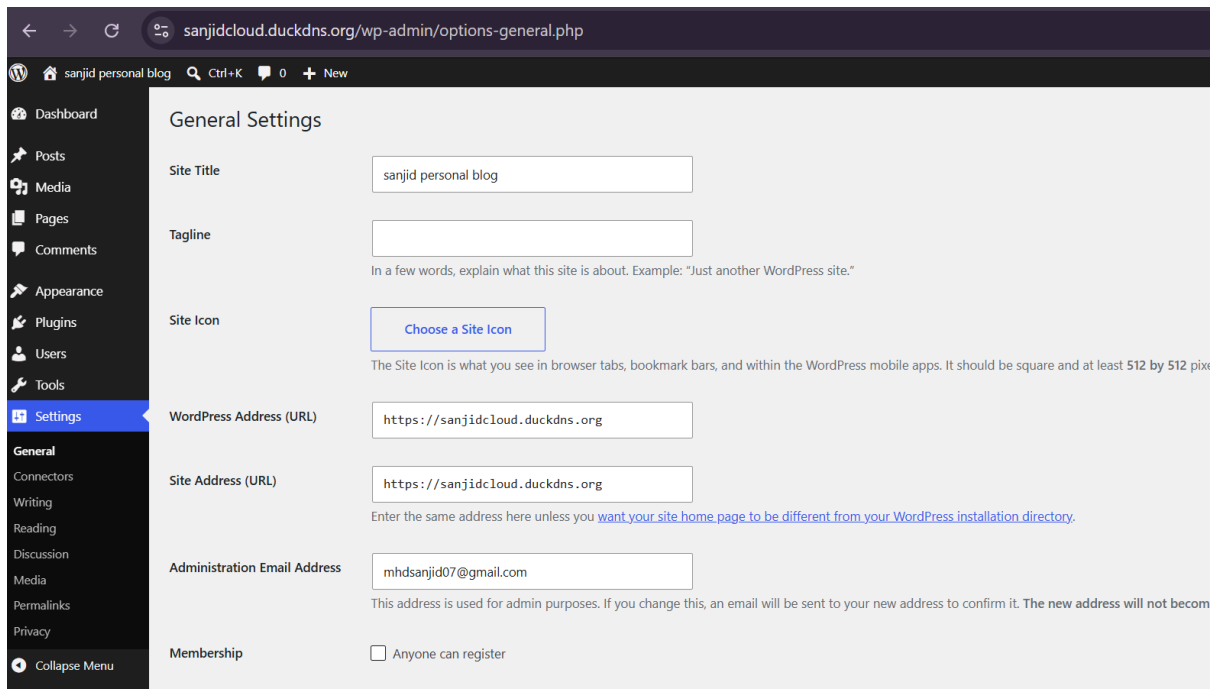
<https://sanjidcloud.duckdns.org/wp-admin/>

The WordPress site settings were configured to use HTTPS.

Both the WordPress Address and Site Address were set to:

<https://sanjidcloud.duckdns.org>

The website title and other basic settings were configured through the WordPress dashboard.



12. Personal Blog Post

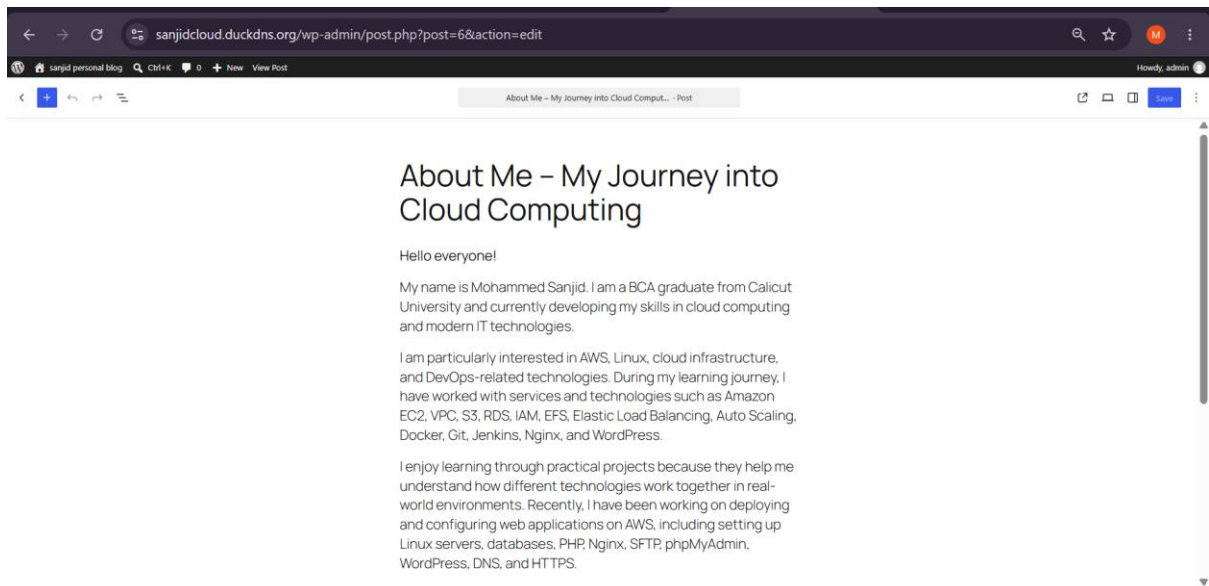
A personal blog post was created through the WordPress dashboard.

The post title was:

About Me – My Journey into Cloud Computing

The post contained information about my educational background, cloud computing interests, and practical experience with AWS and other technologies.

The post was published successfully and was visible on the public website.



Conclusion

The WordPress website was successfully hosted on an AWS RHEL EC2 server. Nginx, PHP, MariaDB, PHP-FPM, SFTP, phpMyAdmin and WordPress were configured successfully. A DuckDNS hostname was configured and a free Let's Encrypt SSL certificate was installed to provide secure HTTPS access. The WordPress dashboard was configured and a personal blog post was created and published successfully.